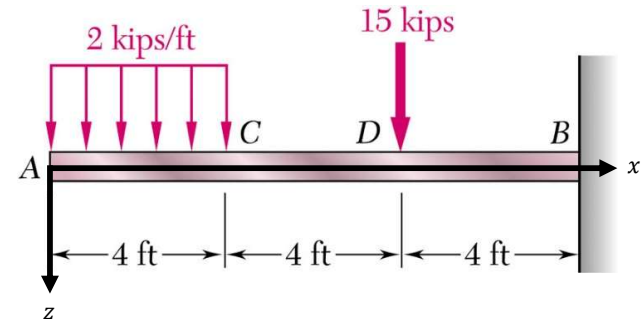


Zeshui Song, 4/5/26, HW7

5.10 Based on free body diagrams, draw shear and bending moment diagrams for the beam with supports and loading shown in the figure below, and determine the maximum absolute value of (a) the shear and (b) the bending moment.



It is also useful to compare this cross-section to standard dimensional lumber sizes.

Given:

- $w = 2 \text{ kips/ft}$

Find:

- The max shear
- The max bending moment

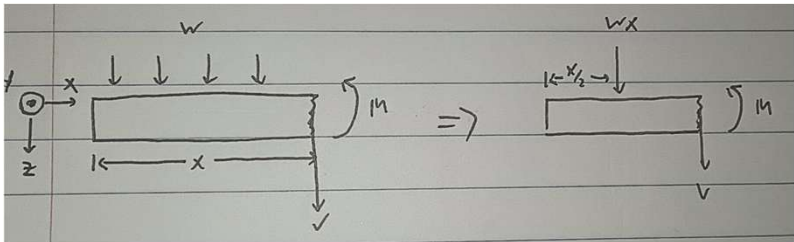
Approach:

- I will analyze the beam separating it into segments, placing "cuts" within each region where the loading remains constant (A-C, C-D, and D-B).
- For each segment, I will use static equilibrium analysis with the +x face sign convention to solve for the internal V and M
- I can then plot the values for V and M across the entire beam and find the maxima.

1

Zeshui Song, 4/5/26, HW7

A to C ($0 < x < 4$)



If we think of the load distribution as a geometric shape, then its line of action is through the *centroid* of that shape. ($x/2$ in this case)

$$\sum F_z = 0 = wx + V$$

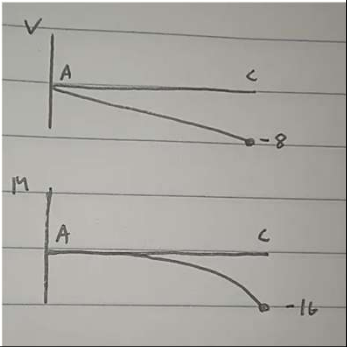
$$V = -wx = -2x \text{ [kips]}$$

Moments about $x=0$

$$\sum M_y = 0 = M - (wx) \left(\frac{x}{2} \right) - V(x)$$

$$M = \frac{w}{2}x^2 + V(x) = \frac{w}{2}x^2 + (-wx)(x) = -\frac{w}{2}x^2$$

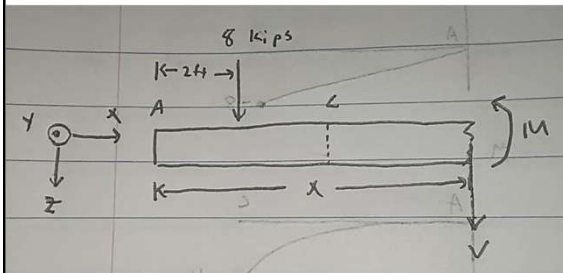
$$M = -\frac{w}{2}x^2 = -x^2 \text{ [kip} \cdot \text{ft]}$$



2

Zeshui Song, 4/5/26, HW7

C to D ($4 < x < 8$)



$$\sum F_z = 0 = -[8 \text{ kips}] - V$$

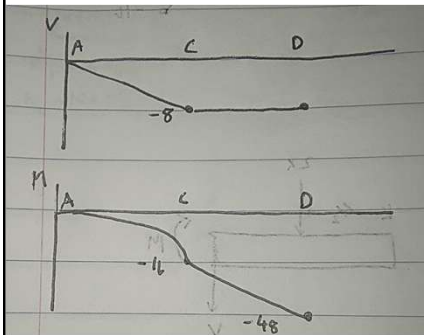
$$\boxed{V = -8 \text{ [kips]}}$$

Moments about $x=0$

$$\sum M_y = 0 = M - [8 \text{ kips}][2 \text{ ft}] - V(x)$$

$$M = [8 \text{ kips}][2 \text{ ft}] + V(x) = [16 \text{ kip} \cdot \text{ft}] + [-8 \text{ kips}]x$$

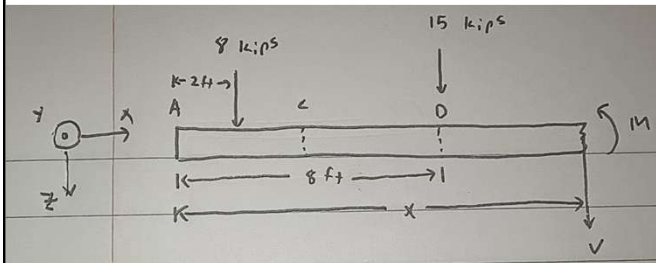
$$\boxed{M = 16 - 8x \text{ [kip} \cdot \text{ft]}}$$



3

Zeshui Song, 4/5/26, HW7

D to B ($8 < x < 12$)



$$\sum F_z = 0 = [8 \text{ kips}] + [15 \text{ kips}] + V$$

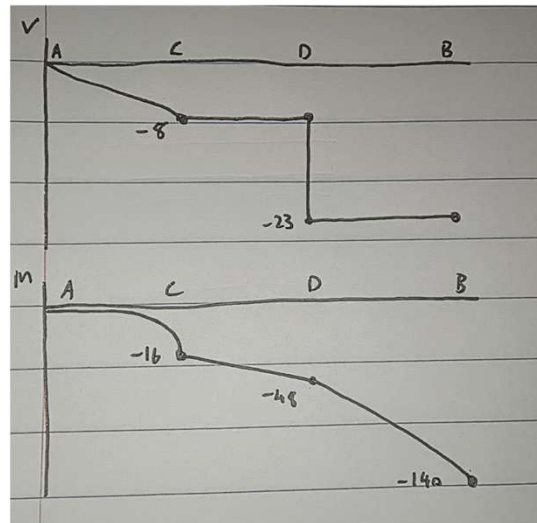
$$\boxed{V = -23 \text{ [kips]}}$$

Moments about $x=0$

$$\sum M_y = 0 = M - [8 \text{ kips}][2 \text{ ft}] - [15 \text{ kips}][8 \text{ ft}] - V(x)$$

$$M = [136 \text{ kip} \cdot \text{ft}] + V(x) = [136 \text{ kip} \cdot \text{ft}] + [-23 \text{ kips}]x$$

$$\boxed{M = 136 - 23x \text{ [kips} \cdot \text{ft]}}$$



4

Zeshui Song, 4/5/26, HW7

From the plot, the max values are:

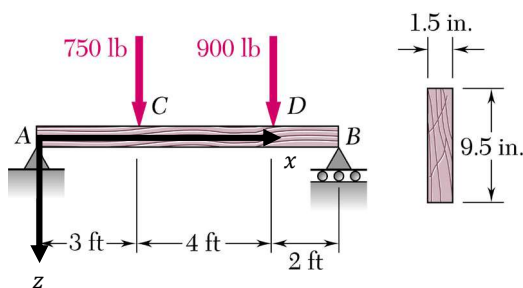
- $M_{max} = -140 \text{ kip} \cdot \text{ft}$
- $V_{max} = -23 \text{ kips}$

This makes sense since the max values occur at the fixed support at B. The directions also make sense since taking moments about B, the moments due to the loads will be CCW, thus requiring a negative CW moment to counter it. [Time spent: 40 mins]

5

Zeshui Song, 4/5/26, HW7

5.15 For the beam and loading show below, determine the maximum normal stress due to bending on the transverse section at C.



Given:

- Two loads of 750 lbf and 900 lbf at C and D

Find:

- Max normal stress due to bending at C

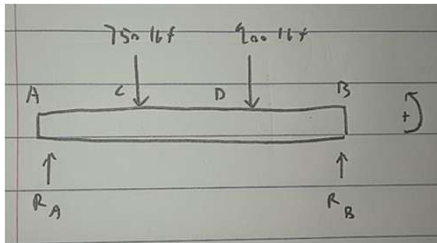
Approach:

- Start by finding the reaction force at A
- Working rightwards from A, find the internal bending moment at C (through a FBD of section AC)
- Use the following equation to find the max normal stress due to bending at C (since the NA is symmetrical, both compressive and tensile max is equal):

$$\sigma_m = \frac{Mc}{I}$$

6

Zeshui Song, 4/5/26, HW7

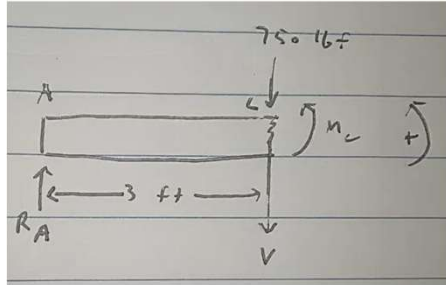


Moments about B to find R_A

$$\sum M_y = 0 = [750 \text{ lbf}][6 \text{ ft}] + [900 \text{ lbf}][2 \text{ ft}] - R_A[9 \text{ ft}]$$

$$R_A = 700 \text{ lbf}$$

At C ($0 < x < 3$)



Moments about C to find M_C

$$\sum M_y = 0 = M_C - R_A[3 \text{ ft}]$$

$$M_C = 2100 \text{ lbf} \cdot \text{ft}$$

$$M_C = 25200 \text{ lbf} \cdot \text{in}$$

7

Zeshui Song, 4/5/26, HW7

Moment of inertia

$$I = \frac{1}{12}bh^3 = \frac{1}{12}[1.5 \text{ in}][9.5 \text{ in}]^3 = 107.171 \text{ in}^4$$

Max normal stress

$$\sigma_m = \frac{Mc}{I} = \frac{[25200 \text{ lbf} \cdot \text{in}][4.75 \text{ in}]}{[107.171 \text{ in}^4]} = 1116.906 \text{ lbf/in}^2$$

This result makes sense since the neutral axis is centered symmetrically, so the compressive and tensile normal stresses will be equal. [Time spent: 20 mins]

8

Zeshui Song, 4/5/26, HW7

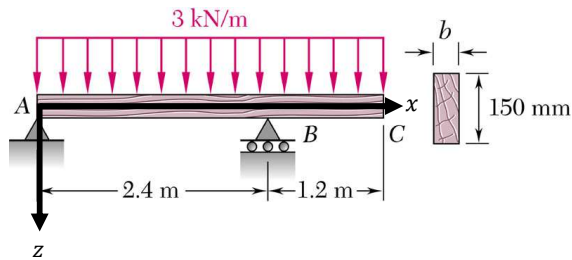
5.69 (Graded) For the uniform distributed load on the beam shown, design the beam cross section, knowing that the grade of timber to be used has an allowable normal stress of 12 MPa. Your work should include sketches showing your coordinate system and showing the distributions of internal shear and moments

Given:

- $w = 3 \text{ kN/m}$
- Width b
- $\sigma_{all} = 12 \text{ MPa}$

Find:

- Width b to not exceed the allowable normal stress.



9

Zeshui Song, 4/5/26, HW7

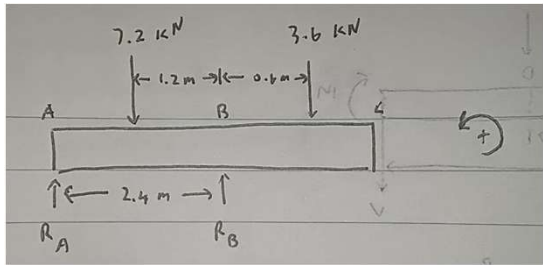
Approach:

- Starting with a moment balance at B to find the reaction forces (R_A and R_B)
- Analyze the beam separating it into segments, placing "cuts" within each region where the loading remains constant (A to B, B to C).
- For each segment, I will use static equilibrium analysis with the +x face sign convention to solve for the internal V and M
- I can then plot the values for V and M across the entire beam and find the maximum bending moment and use the equation below to find the maximum normal stress:

$$\sigma_m = \frac{Mc}{I}$$

Where the moment of inertia will tell me the dimension for b

Moments about B to find R_A and R_B



$$\sum M_y = 0 = -R_A[2.4 \text{ m}] + [7.2 \text{ kN}][1.2 \text{ m}] - [3.6 \text{ kN}][0.6 \text{ m}]$$

$$R_A = 2.7 \text{ kN}$$

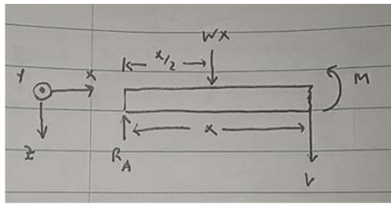
$$\sum F_z = 0 = [7.2 \text{ kN}] + [3.6 \text{ kN}] - R_A - R_B$$

$$R_B = 8.1 \text{ kN}$$

10

Zeshui Song, 4/5/26, HW7

A to B ($0 < x < 2.4$)



$$\sum F_z = 0 = -R_A + wx + V$$

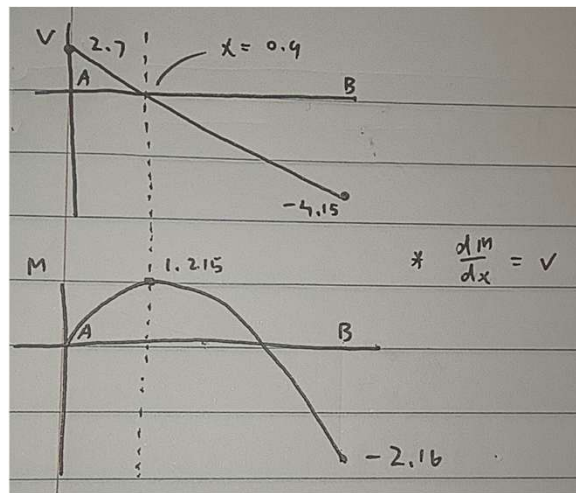
$$V = R_A - wx = 2.7 - 3x \text{ [kN]}$$

Moments about $x=0$

$$\sum M_y = 0 = M - wx \left(\frac{x}{2}\right) - V(x)$$

$$M = \frac{1}{2}wx^2 + V(x) = \frac{3}{2}x^2 + (2.7 - 3x)x$$

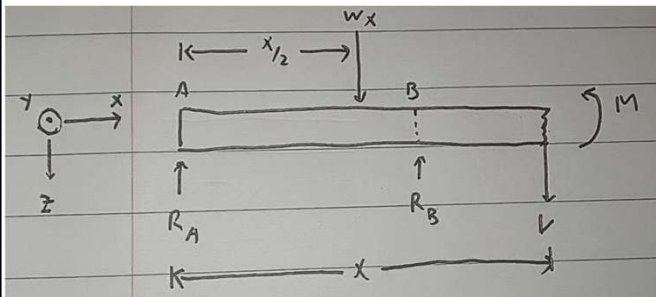
$$M = -1.5x^2 + 2.7x \text{ [kN} \cdot \text{m]}$$



11

Zeshui Song, 4/5/26, HW7

B to C ($2.4 < x < 3.6$)



$$\sum F_z = 0 = -R_A - R_B + wx + V$$

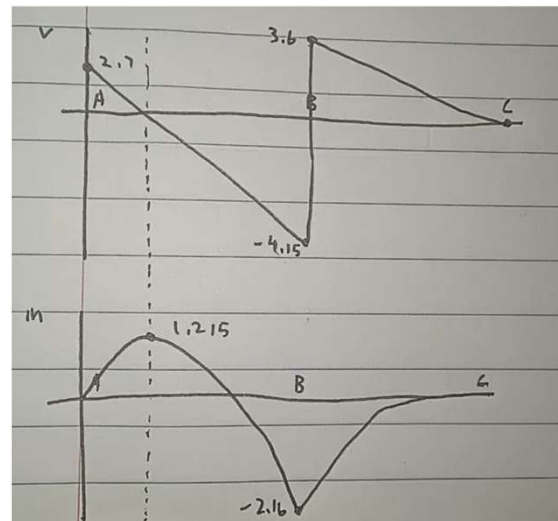
$$V = R_A + R_B - wx = 10.8 - 3x \text{ [kN]}$$

Moments about $x=0$

$$\sum M_y = 0 = M - wx \left(\frac{x}{2}\right) + R_B[2.4 \text{ m}] - V(x)$$

$$M = -19.44 + \frac{1}{2}wx^2 + V(x) = -19.44 + \frac{3}{2}x^2 + (10.8 - 3x)x$$

$$M = -19.44 - 1.5x^2 + 10.8x \text{ [kN} \cdot \text{m]}$$



12

Zeshui Song, 4/5/26, HW7

Moment of inertia

$$I = \frac{1}{12}bh^3$$

Solve for b using max allowable normal stress and the max bending moment $M = -2.16 \text{ kN} \cdot \text{m}$

$$\sigma_{all} = |\sigma_m| = \left| \frac{Mc}{I} \right| = \left| \frac{Mc}{\frac{1}{12}bh^3} \right|$$

$$b = \frac{12|M|c}{\sigma_{all}h^3} = \frac{12[2.16 \times 10^3 \text{ N} \cdot \text{m}][0.075 \text{ m}]}{[12 \times 10^6 \text{ N/m}^2][0.15 \text{ m}]^3} = 0.048 \text{ m}$$

The shear and bending moment distributions make sense because

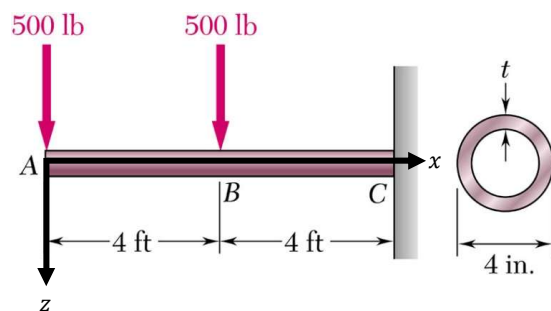
- Max shear occurs at A and B, where the supports are
- Bending moment is zero at C since there are no supports there

The results show that in overhang designs, the supports are the critical regions where failure is most likely to occur. Interestingly, the direction of the bending moment changes, which means the beam bends in multiple directions across its length. [Time spent: 35 mins]

13

Zeshui Song, 4/5/26, HW7

5.82 A steel pipe of 4" diameter will support the two loads shown below. Knowing that the stock of pipes available has wall thicknesses varying from 1/4" to 1" in 1/8" increments, and that the allowable normal stress for the steel used is 24 ksi, determine the minimum wall thickness t that can be used.



Given:

- $d_o = 4 \text{ in}$
- $\sigma_{all} = 24 \text{ ksi}$

Find:

- Minimum wall thickness t (t is in increments of 1/8 in from 1/4 in to 1 in)

Approach:

- Starting from A and working rightward, use FBD sections (A-B and B-C) to find the max bending moment across the pipe
- Find the max normal stress:

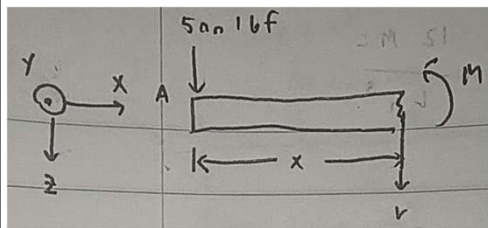
$$\sigma_m = \frac{Mc}{I}$$

- Use the max allowable stress to find min thickness

14

Zeshui Song, 4/5/26, HW7

A to B ($0 < x < 4$)



$$\sum F_z = 0 = [500 \text{ lbf}] + V$$

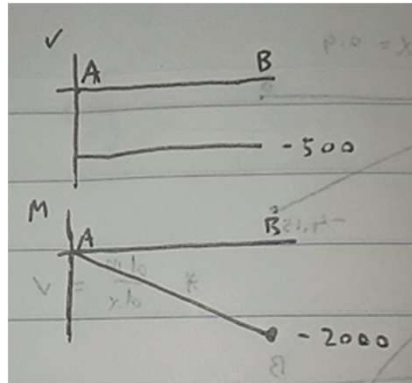
$$\boxed{V = -500 [\text{lbf}]}$$

Moments about $x=0$

$$\sum M_y = 0 = M - V(x)$$

$$M = V(x) = (-500)x$$

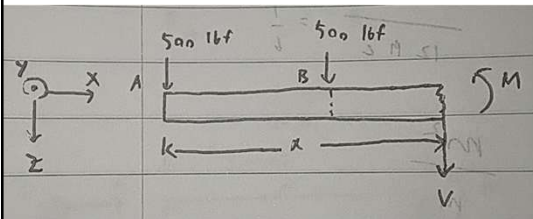
$$\boxed{M = -500x [\text{lbf} \cdot \text{ft}]}$$



15

Zeshui Song, 4/5/26, HW7

B to C ($4 < x < 8$)



$$\sum F_z = 0 = [500 \text{ lbf}] + [500 \text{ lbf}] + V$$

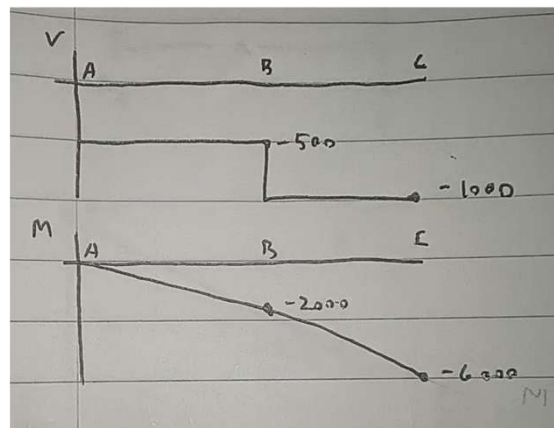
$$\boxed{V = -1000 [\text{lbf}]}$$

Moments about $x=0$

$$\sum M_y = 0 = M - [500 \text{ lbf}][4 \text{ ft}] - V(x)$$

$$M = [2000 \text{ lbf} \cdot \text{ft}] + V(x) = [2000 \text{ lbf} \cdot \text{ft}] + (-1000)x$$

$$\boxed{M = 2000 - 1000x [\text{lbf} \cdot \text{ft}]}$$



16

Zeshui Song, 4/5/26, HW7

Moment of inertia

$$I = \frac{\pi}{64} (d_o^4 - d_i^4)$$

Max normal stress

$$\sigma_{all} = |\sigma_m| = \frac{|M|c}{I} = \frac{|M| \left(\frac{d_o}{2} \right)}{\frac{\pi}{64} (d_o^4 - d_i^4)}$$

$$d_i = \sqrt[4]{d_o^4 - \frac{32|M|d_o}{\pi\sigma_{all}}} = \sqrt[4]{[4 \text{ in}]^4 - \frac{32[6000 \cdot 12 \text{ lbf} \cdot \text{in}][4 \text{ in}]}{\pi[24 \times 10^3 \text{ lbf/in}^2]}} = 3.400 \text{ in}$$

$$t = \frac{d_o - d_i}{2} = 0.2995 \text{ in}$$

The closest 1/8 in increment that is larger than 0.299 in is: $t = \frac{3}{8} \text{ in}$

This result shows that hollow tubes are actually quite efficient under bending moments since the max stress is at the surface so the inner region can be hollow.
[Time spent: 30 mins]